

AI And IoT: Redefining SEMICONDUCTOR DEVELOPMENT

Uncover the driving forces behind the semiconductor revolution—how AI slashes human error, unearths untapped revenue streams, and underscores the pivotal role of customisation with domain-specific chipsets



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The surge in demand for artificial intelligence (AI) and the Internet of Things (IoT) devices is reshaping how we interact with technology and the world around us. From smart trackers in our watches to intelligent thermostats in our homes, the application of artificial intelligence (AI) is permeating every aspect of our lives. A consumer survey predicts that by 2030, we will be surrounded by more than 350 billion AI IoT devices, underlining the rapid growth in demand for these devices.

While the proliferation of AI IoT devices is impressive, it also raises questions about the role of AI in improving our lives and the potential for human error in these technologies. In this article, we will explore the key trends in AI applications, the need for increased computational power, and the driving agents behind this revolution in the semiconductor industry.

AI applications and reducing human error

The primary goal of integrating AI into IoT devices is to enhance their capabilities and reduce human error. AI can provide more accurate results than the human eye or persistence in various applications, such as image recognition. For instance, thanks to machine computing power, Google's image recognition application significantly reduced human error rates when AI was utilised. As a result, there is a growing need for complex system-on-chip (SoC) architectures to support these AI use cases. These architectures must evolve to accommodate single-threaded applications, OpenCL support, benchmarking, multi-threaded networking applications, and image recognition.

In networking applications, focus lies in leveraging network function processors to enhance system performance. One significant aspect that has been emphasised is security, primarily due to the vast data transmitted between edge devices and cloud services. We must ensure the integrity and authenticity of this data along its journey.

We are actively implementing robust computing mechanisms within the communication channel to achieve this. These mechanisms are designed to verify and validate

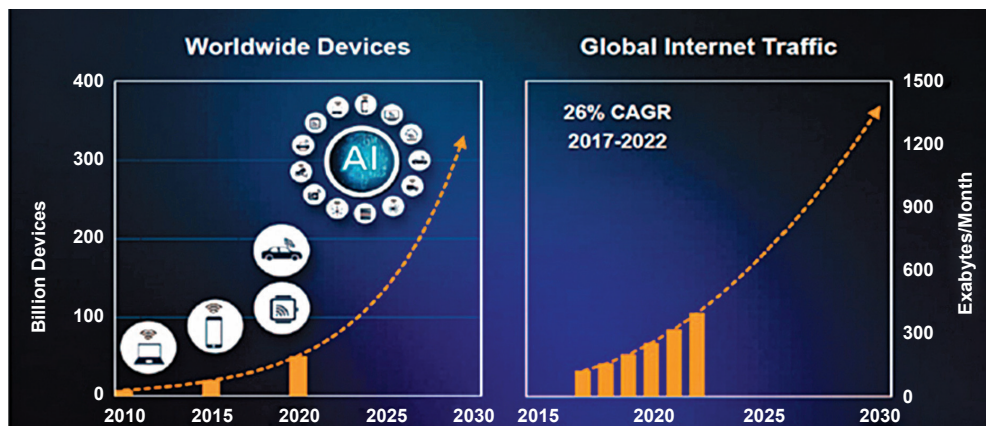


Fig. 1: More than 350 billion AIoT (AI+IoT) devices in year 2030 (Source: 2020 IEEE ISSCC, Cisco VNI Global IP Traffic Forecast 2017-2022)